

Introduction

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Python Introduction

- *TIP: If viewing this in a web browser, press **ESC** to see a preview of all the slides.*
- Python is an *interpreted* language
- Statements can be interactively ran and tested
- 4 Ways to do this:
 1. The Python Interpreter
 2. IPython Interpreter
 3. Self-Contained Scripts
 4. Jupyter Notebook

The Interpreter

- Access by typing `python` in a command prompt. In this form you can type statements and execute code
- This places you in the Python **REPL**
 1. Read the input
 2. Evaluate it
 3. Print the results
 4. Loop back to step 1
- `>>>`
- `help()` and Python reference

IPython Interpreter

- A more advanced and feature packed version of the python interpreter.
- IPython will use numbered commands in place of the
>>>
- Input and Output are numbered.

Self-Contained Scripts

- Statements can be saved to a file, much like other programming languages (C++, Java)
- Python scripts are saved in `.py` files
- To run a script, assuming you are in the same directory as the file:
 - `python [file name].py`

The Jupyter Notebook

Hybrid of the two environments that will be explored and explained later in the course.

The Python Syntax

- *Syntax* - The structure of the language (incase you forgot).

Example program

```
# set the midpoint
```

← Comment

```
midpoint = 5
```

← Variable Assignment

```
# make two empty lists
```

```
lower = []; upper = []
```

```
# split the numbers into lower and upper
```

```
for i in range(10):
```

```
→ if (i < midpoint):
```

```
→ lower.append(i)
```

```
else:
```

```
upper.append(i)
```

lower = []
upper = []

```
print("lower:", lower)
```

```
print("upper:", upper)
```

if
else if
else

if
elif else:

for (int i=0; i<10; i++)
{

cout << i;
if (i < 10)

Syntax Rundown

- Comments are marked by “#”
- End-Of-Line Terminates a Statement
- Semicolons can be used optionally used to terminate

Syntax Rundown (Continued)

- Indenting: Whitespace matters
 - Code blocks are controlled by indents
 - 4 spaces is the preferred amount
- Whitespace *within* a line doesn't matter
- Parentheses are for grouping or calling

Variables

- Similar to others
- Use the assignment operator
- `x = 4` assigns a value of 4 to the variable `x`

Note on Variables

- Python is *dynamically-typed*
 - Opposed to a static-typed language like C++
- Variables are *pointers* to space in memory
- You can do this:
 - `x = 1`
 - `x = 'hello'`
 - `x = [1, 2, 3]`

More Notes

- Python simple types (int, float) are *immutable*
- Lists on the other hand are *mutable*
- Example:
 - `x = 10` x points to a memory location containing the value 10
 - `x = 11` x points to a new memory location with the value 11.

Everything is an Object

- `type()` function
- Objects will have a type

What is an Object?

- In terms of programming languages an *object* is:
 - an entity that contains data along with metadata and/or functionality.
- Python Objects will:
 - have metadata (better known as *attributes*)
 - have associated functionality (*methods*)
- dot syntax